Figure 17.10 Solving TSP for a sub-graph of the K_{10} graph shown in figure 17.8. The figure highlights the K_6 complete graph formed by the cities belonging to the DC Universe, and the best solution found. The rest of the vertices/edges of the original graph are grayed out for clarity

To provide a clearer view and description of the process, we need to keep the set of cities small: for instance, we can restrict to the 6 DC Universe cities in figure 17.8, obtaining the graph shown in figure 17.10, for which there are just 720 possible permutations of the vertices – which produces the landscape shown in figure 17.11, which is far less clogged than what we had shown in figure 17.9.

Now, since we have the full landscape, you might point out that we could just brute-force search it and find the best solution: and you'll be right, we have evaluated the cost of each single permutation to draw the chart in figure 17.11, so we can just extract its minimum: as shown in the figure, the best solution is the sequence:

```
[New Carthage, Happy Harbor, Gotham City,
Metropolis, Opal City, Civic City]
```

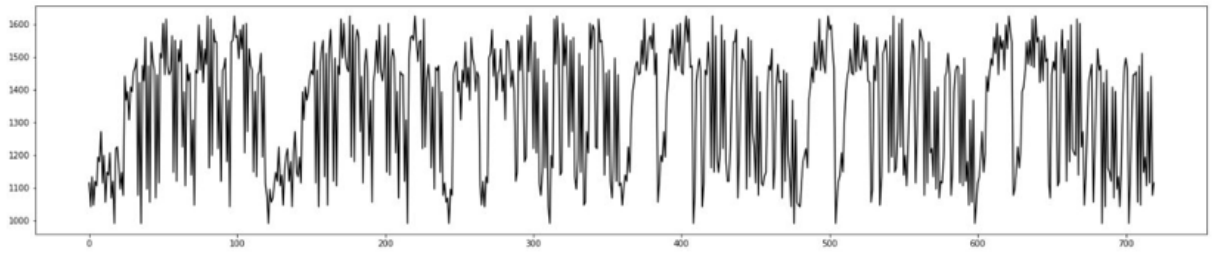


Figure 17.11 The cost landscape for TSP applied to the complete graph K6 as shown in figure 17.10.